

Falling for phishing attempts: An investigation of individual differences that are associated with behavior in a naturalistic phishing simulation

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Declaration of interest:

Funding sources: The work has been supported by the Cyber Security Research Centre Limited whose activities are partially funded by the Australian Government's Cooperative Research Centres Programme.

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Highlights

- We investigated individual differences in cybersecurity behavior in a naturalistic phishing simulation
- We found that lower phishing email detection accuracy only modestly predicted riskier cybersecurity behavior
- Instead, fewer employment years and lower employee satisfaction and loyalty were most predictive of riskier behavior
- These results suggest that newer employees are more vulnerable to phishing attempts
- Furthermore, interventions that increase employee satisfaction might be effective at reducing risky cybersecurity behavior

Abstract

Social engineering cyber-attacks such as phishing emails pose a serious threat to the safety of many organizations. Given that the effectiveness of these attacks heavily relies on poor human decision making, an improved understanding of the individual characteristics that increase cybersecurity vulnerability could inform more targeted training. The current study aimed to identify whether several factors, including phishing email detection ability, confidence in one's phishing identification decisions, attitudes towards one's level of responsibility and efficacy, and employee satisfaction and loyalty to the organization, can predict behavior in a naturalistic phishing simulation in an employment setting. We followed up employees of a large organization who had been recently targeted by a phishing simulation and asked them to complete a survey that included a phishing detection task. The employees' behavior in the phishing simulation was ranked according to its safety: reporting the suspicious email, neither reporting nor clicking on the embedded link, and clicking on the link. We found that fewer years of employment at the organization and lower employee satisfaction and loyalty predicted increasingly unsafe behavior in the simulation. This suggests that newer and unsatisfied employees are most vulnerable to phishing attempts and might benefit most from targeted cybersecurity training.

Keywords: phishing, cybersecurity behavior, naturalistic simulation, detection accuracy, attitudes

Introduction

Social engineering cyber-attacks such as phishing emails pose a serious threat to the safety of company finances, intellectual property, and other sensitive assets. In 2019, 1.45 trillion USD was lost to cybercrime (RiskIQ, 2019), a figure that is projected to increase by a factor of 10 in the next five years (Morgan, 2020), and Verizon (2020) estimates that stock corporate prices drop by 5% after such a breach. Of these direct costs, approximately 10% is accounted for by phishing attacks (RiskIQ, 2019), however, the relative frequency and success, and therefore cost, of phishing attacks is projected to increase exponentially (FBI, 2020). Whereas more sophisticated methods of cybersecurity attack rely on limitations or manipulation of software, the likelihood of phishing attacks being effective relies heavily on poor human decision making.

Given this, an improved understanding of the individual characteristics that increase cyber-security vulnerability is highly desirable since it could inform more targeted training. Thus, the importance of precise and thorough articulation of the pertinent cognitive and behavioral processes that contribute to that risk is clear. Previous research into the human factors and cognitive processes involved in phishing susceptibility has largely focused on the content messaging and sentiment or philological components of the most successful phishinges (Aggarwal et al., 2014; Avery et al., 2017; Pent et al., 2018; Verma et al., 2012), the visual elements that can be used to identify phishinges (Dewan et al., 2017; Jain & Gupta, 2017; Medvet et al., 2008), basic demographics and employment characteristics associated with identified risk (Asfoor et al., 2018; Lain et al., 2021), computer expertise (Williams et al., 2017), estimation of risk (Corradini & Nardelli, 2018), and other organizational elements such as allocated resources, workload, and knowledge and awareness information (see Desolda et al., 2021, for a recent systematic review). Although these variables can be useful for individuals to orient to known indicators of phishinges and, therefore, potentially inform

1 organizational, cybersecurity training, and for organizations to identify those most at risk
2 based on variables that cannot be changed, such as sex and age, investigation into
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4 cybersecurity-related behaviors should focus on identifying processes that allow for targeted
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6 behavioral training based on variables that can be improved or manipulated. Further, the
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8 majority of existing research has been conducted in lab environments, where the risk of being
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10 phished is not meaningful, and where workload or social-organizational factors are
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12 minimised. Research should, therefore, seek to instantiate mitigated-risk real-world simulated
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14 phishes in organizational contexts to better understand the effects of organizational, social,
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16 and personal susceptibility to being phished.
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22 Many large organizations use various forms of educative tools and training programs
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24 to minimise the risk of falling victim to cybersecurity attacks, with simulated phishing attacks
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26 deployed via email to employees becoming increasingly common (Jayatilaka et al., in prep.).
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28 Nevertheless, the efficacy of such tools is often assumed rather than tested, and the
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30 mechanisms by which that putative efficacy might operate are not known. If the mechanisms
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32 by which these tools could potentially improve behavior by some means are not known, then
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34 it is unclear whether generic tools deployed to all employees of an organization are a time-
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36 and cost- effective, if indeed effectual, approach, or whether training targeted to an
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38 identifiable locus of risk would yield better outcomes. In addition to this, the extent to which
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40 any metrics yielded by phishing simulation campaigns map onto, that is, provide a veridical
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42 account of, actual engagement with non-simulated attacks is certainly pertinent and not yet
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44 known.
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51 Many organizations assess employee vulnerability to phishing attacks by performing
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53 regular blinded phishing simulations in which employees are sent emails that contain cues
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55 that could be used to identify phishing attacks (i.e., suspicious sender URLs, spelling
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57 mistakes; e.g., Abawajy, 2014; Dodge et al., 2007; Landesman, 2016; Williams et al., 2018).
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1 The measures yielded by current phishing simulations are click rate and reporting rate. Click
2 rate reflects the proportion of successful phishing attempts (i.e., errors in judgement where a
3 link in an illegitimate email is clicked on by an employee). Reporting rate reflects the
4 proportion of simulated phishing links that were reported using the organization's reporting
5 process for suspected phishing emails.
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11 These approaches yield a crude measure used to assess susceptibility to being phished
12 (click-rate) and organizational risk per employee (a combination of click rate and reporting
13 rate) and allows an organization to monitor performance and changes in performance
14 between departments and within employees. That is, phishing simulations provide an overall
15 metric of performance, but this metric is relatively uninformative of the causes of that
16 performance. The content and type of training or remediation that is implemented to
17 employees identified as susceptible should be informed by those causes so it can effectively
18 improve performance based on the causes of its deficit. So, this approach cannot deliver
19 meaningful training that will improve performance because it is not individualised based on
20 the true behavioral or cognitive deficits in employees, and, thus, minimising costly employee
21 behaviors requires a more in-depth approach.
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39 This approach is nevertheless useful as it allows the identification of employees who
40 fall victim to phishing attacks in their natural online environment. However, while this
41 method identifies both individuals who fall prey to the simulated phishing attack ('clickers')
42 and individuals who accurately perceive the email and report it ('reporters'), a majority of
43 individuals do neither. That is, approximately half of individuals neither click on the link in
44 the phishing email nor do they report it. This is problematic, as it is impossible to determine
45 what might cause this behavior, which limits the organization's ability to anticipate what kind
46 of training might benefit this majority of individuals. In principle, several factors might be
47 responsible for this behavior, reviewed in more detail below.
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Identification

These individuals might lack the ability to identify the email as a phishing attempt (hence they did not report it) and they may not have clicked on the link because they did not open the email or disregarded it for other reasons. Overall click rate (i.e., an employee clicking a phishing email link) is assumed to reflect failure to identify the link as illegitimate. However, according to signal detection theory, click rate is potentially confounded by an individual's bias toward or away from perceiving an email as a phishing threat, and reflects the outcome of a combination of this bias and their true ability to discriminate between legitimate and phishing emails. Signal detection theory has been used to measure the relative contribution of these processes to the detection of a target in a series of otherwise irrelevant items (Green & Swets, 1966; Syanislav & Todorov, 1999), and the parameters yielded by signal detection analysis have been shown to map onto neural behavior (Heeger & Landy, 1997; Tyler & Chen, 2000), confirming its utility in probing underlying processes of neurocognition. According to this model then, the identification of a phishing target is the product of discrimination (or *sensitivity*, d' - in this context, the ability to differentiate between phishing attempts and legitimate emails) and bias (or *decision criterion*, C - in this context, the tendency to treat all emails as potential phishing attempts).

Signal detection theory considers all behavioral patterns, which provides the benefit of additional insight and understanding of behavior based on all potential response behaviors (see Figure 1). That is, it can allow an organization to understand whether an employee requires additional training at identifying suspicious emails or whether an employee is more or less likely to perceive an ambiguous email as suspicious based on a bias that is the result of perceived risk of error. For instance, suppose an ambiguously suspicious email is sent to a pool of 100 employees. These employees, thus, receive the same information (e.g., a friendly greeting, a masked URL, a misspelled word) and, in their natural employment environment,

are tasked with approaching this email as any other. There are four possible outcomes (shown in Figure 1). Either there is a phishing attempt present or there is not, and the employees will identify (either correctly or incorrectly) the email as a phishing attempt or they will not. Using such a matrix, signal detection theory allows an organization to determine whether a failure to detect a phishing target (i.e., clicking the malicious link in a phishing email) is the result of poor detection ability or a liberal risk bias (a tendency to assume emails are legitimate). Quantifying such a process could feasibly inform the type of training that will be effective for a given employee. That is, if a given employee falls victim to a phishing attack, many organizations deploy generic training that aims to improve detection ability; if, however, this employee has adequate detection ability (indicated by a high value of d'), and their clicking was the result of a bias away from the phishing decision boundary (indicated by a negative C value), then such training will be ineffective and potentially even have negative effects on subsequent cybersecurity behavior and engagement (e.g., some evidence suggests that negative perceptions of training increases turnover and reduces motivation to adhere to organizational protocols and employee loyalty; Ashar et al., 2013; Dysvik & Kuvaas, 2008; Newman et al., 2011; Rawashdeh & Tamimi, 2019). Signal detection theory is an established method in phishing detection tasks (see Butavicius et al., 2016; Canfield & Fischhoff, 2018; Canfield et al., 2016; Martin et al., 2018; Lawson et al., 2020), but it is yet to be determined whether (1) the parameters yielded by the analysis map onto organizational or social variables, or (2) whether training based on those parameters can improve performance, that is, whether the parameters are sufficiently specific to discriminate between types of performance in a realistic simulation.

Figure 1

Matrix of possible instances when receiving and actioning an email.

	Response “yes”	Response “no”
Phish present	Hit	Miss
Phish absent	False alarm	Correct rejection

Confidence in decision

The extent to which one rates the accuracy of their own conclusions may contribute further to reporting behavior. Research has shown that confidence in one’s judgements affects the likelihood of performing a task when that task is optional (Ahmad, 2020; Heath & Tversky, 1991; Johnson et al., 2011). That is, underconfidence in one’s judgement that a given email is a phishing attack could hypothetically reduce the likelihood of reporting. An employee might therefore have accurately detected the phishing attempt (and hence did not click on the link) but they may have lacked the confidence in their ability and did not report it because they were unsure this was a phishing attempt. Whereas, on the other hand, overconfidence might under some conditions lead to overreporting; overreporting may be associated with increased cost due to observation and potentially information loss due to the removal of legitimate emails from organizational circulation.

In any case, poor confidence calibration, that is, incongruence between an individual’s actual accuracy and their perceived accuracy, may increase the probability of legitimate emails being flagged as phishes, and reduce the likelihood of phishing emails being reported. An organization may choose to reinforce over-reporting or under-reporting depending on their associated costs: over-reporting might be more cost-effective than under-reporting if successful phishing attempts are associated with large losses; alternatively, over-reporting might be less cost-effective if the cost of investigating too many potential threats may outweigh the risk associated with failure to identify or report phishes. Regardless, from a

practical perspective, training that helps employees calibrate their confidence rather than simply overtraining identification ability, specifically in instances where identification ability is sufficient in the presence of underconfidence, might be most beneficial.

Attitudes toward the organization

Thus, employee attitudes toward the organization and its training procedures may influence behaviors related to the behaviors associated with that training, that is, cybersecurity. High employee satisfaction has a well-established positive effect on various domains of organizational performance (Argyle, 1989). Likewise, employee loyalty is known to exert a positive influence on some behaviors, including adhering to some organizational procedures and protocols (e.g., Guillon & Cezanne, 2014; Yee et al., 2010). Whether job satisfaction and organizational loyalty exert such positive effects on effort in terms of in identifying phishes or in reporting suspicious emails seem equally plausible, but this effect has not been studied. Indeed, if overtraining or perceived overregulation of behavior in the workplace contribute to poorer job satisfaction or organizational loyalty (Ashar et al., 2013; Dysvik & Kuvaas, 2008; Newman et al., 2011; Rawashdeh & Tamimi, 2019), the relative impact of inappropriate training, and the relationship between training, signal detection parameters of performance in those who fall for phishes and those who do not, and job satisfaction and organizational loyalty are important to investigate.

Attitudes toward reporting

While insights from signal detection theory and organizational attitudes can help us understand behaviors associated with click rate, reporting suspicious emails is an equally important cybersecurity behavior as it reduces the reach, and therefore potential risk, of such suspicious emails. So, whereas attitudes about the job and organization may contribute to phish identification, attitudes about oneself and the extent to which one perceives one's actions to exert tangible outcomes may contribute to reporting behavior. For instance, an

1 employee might have detected the phishing attempt but did not report it due to a suboptimal
2 attitude towards reporting (e.g., lack of responsibility, moral disengagement, or lower self-
3 efficacy).
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7 Actions requiring even minimal effort that relate in some way to a directive, moral
8 ambiguity, uncertain outcome, or where the action relies to some extent on skill, knowledge,
9 or a judgement rely on the internalisation of responsibility for such actions and for the
10 negative outcomes of either engaging in or avoiding such actions (Bandura et al., 1996; see
11 also Moore, 2015). Bandura and colleagues (1996) articulated eight strategies that displace
12 the internalised responsibility of committing these actions, which each plausibly influence the
13 likelihood that an employee will internalise the responsibility of reporting a phishing email,
14 or internalise the risk and potential negative outcomes as attributable to their action or lack
15 thereof, instead of externalising attribution to their colleagues. Diffusion of responsibility,
16 one of the eight strategies identified by Bandura (1996), is known to lead to a reduction in
17 actions where responsibility for taking an action is shared by all individuals of a group, such
18 as a large organization (Hiekkataipale, & Lämsä, 2019; West, 2000). The mechanism by
19 which diffusion or displacement of responsibility is thought to operate is by minimising
20 outcome monitoring, and the effect is even observed when a negative outcome is the direct
21 result of one individual if another individual, who could not be responsible for the outcome,
22 is present (Beyer et al., 2017).
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46 If externalising attitudes do indeed predict lower likelihood of reporting behavior,
47 then it is unlikely that training or educational tools that highlight the impact of phishing will
48 be effective, since taking actions to protect against that impact is not considered to be the
49 responsibility of the employee. Employees identified to have high diffusion of responsibility
50 and low internalised moral engagement could potentially be trained to remediate these
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1 recalcitrant behaviors, however evidence of whether these attitudes can be meaningfully
2 manipulated is mixed (see Scully & Rowe, 2009).
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4 On the other hand, an employee who internalises the responsibility of engaging in
5 simple behaviors that could reduce organizational risk might be expected to engage those
6 behaviors only if they believe their behaviors truly exert a positive, protective effect. Indeed,
7 if an employee feels that reporting an email as suspicious will not yield a benefit to the
8 organization by reducing exposure of that email to their colleagues or some other mechanism,
9 it is unlikely that they will report the email. Evidence from health psychology reliably shows
10 that the extent to which a person engages in health-promoting behaviors is mediated by their
11 self-efficacy, that is, the extent to which they believe they have the capacity to engage in
12 those behaviors, and the extent to which they believe those behaviors will actually confer a
13 benefit on their health. This effect has been reported in numerous domains, but not yet in
14 cybersecurity behavior.
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17 **The present study**

18 Here, we intend to identify the factors associated with phishing susceptibility and
19 phishing email reporting, namely, (1) the ability to discriminate real and malicious emails, (2)
20 confidence in discriminating real and malicious emails and/or (3) attitude-related factors. The
21 results of this experiment are likely to improve our understanding of the causes of different
22 behaviors observed in the phishing simulation, which may facilitate the development of
23 phishing interventions that target employee-specific vulnerabilities. Currently, it is unclear
24 whether the behavior of individuals who click on simulated phishing links is solely due to a
25 reduced ability to identify phishing attempts, or why individuals who do not report a phishing
26 email nor click on the link behave in this way, and this lack of knowledge prevents the
27 organization from taking steps to improve these individuals' behavior. In principle, these
28 individuals might differ from reporters not only in terms of their ability to detect phishing
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1 emails, but also in their confidence in their ability to detect phishing attempts, or attitudes
2 that are conducive to more cautious or reporting behavior. The results of this study could
3 provide important clues for designing interventions that will improve reporting rates in these
4 individuals. For example, if we find that these individuals are generally able to discriminate
5 phishing from non-phishing emails but their confidence in their own ability is not well
6 calibrated (that is, they lack confidence in their decisions which likely results in reduced
7 reporting rates), then training that relies heavily on teaching individuals to recognise the
8 extent of their abilities via experience with explicit feedback might help them calibrate their
9 confidence so that it better matches their performance level, which would likely improve their
10 reporting rates. Alternatively, if we find that these individuals are able to discriminate
11 between phishing and non-phishing emails and are appropriately confident in their decisions,
12 but their survey answers suggest they lack a sense of corporate responsibility or believe their
13 actions would not have a significant effect on cybersecurity, then the best approach to
14 improve their reporting rates would likely be via interventions that aim to change employee
15 attitudes rather than further phishing training. Understanding the reasons behind similar
16 behaviors would therefore guide the choice of intervention that is most likely to result in
17 behavioral change.

41 Method

43 Sampling procedure and participants

45 Employees of a large Australian financial organization were invited via email to
46 participate in the experiment. The email was sent by the organization's cybersecurity team to
47 employees' workplace email addresses. As part of its cybersecurity training and assessment,
48 this organization runs monthly phishing simulations on approximately 4,000 randomly-
49 selected employees. The randomly chosen employees are 'phished' in an initial campaign and
50 their behavior is recorded (clicking on the phishing link, reporting the email containing the
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phishing link as unsafe/suspicious, or neither clicking nor reporting). Those who clicked on the link in the initial campaign receive brief training, and are later followed up in a similar campaign in which their behavior is again recorded. Those who report the suspicious email receive a thanking email, but are not followed up with another campaign, and neither are those who did nothing in the initial campaign. This yields five possible groups denoted by their behavior in the initial and follow-up campaigns (separated by a slash): *report/-*, *nothing/-*, *click/report*, *click/nothing*, and *click/click*.

With the exception of having been targeted by the most recent phishing simulation campaign, there were no other inclusion criteria as the purpose of this study was to sample employees representative of both the organization's broad departmental structure and categories of behavior in previous phishing simulation campaigns. Invitations to participate in the study were sent in August and September 2021. Of 235 participants who started the experiment, $N = 138$ (65 women, 71 men, 2 unspecified; aged 42.27 ± 10.38 years) completed at least the phishing discrimination task (64 *report/-*, 25 *nothing/-*, 23 *click/report*, 19 *click/nothing*, and 7 *click/click*). Of these, $N = 126$ provided gender information and also completed all self-reported measures (59 women and 67 men; aged 42.22 ± 10.34 years; 60 *report/-*, 20 *nothing/-*, 21 *click/report*, 18 *click/nothing*, and 7 *click/click*).

Procedure

The invitation email included a link which sent participants to a protected Qualtrics (Qualtrics, UT, USA, 2020) questionnaire. Participants were made aware that participation was voluntary and that they could not be identified by their employing organization, and, although participants were not directly reimbursed for their participation, the protocol was administered during business hours. After giving informed consent, participants were administered a short questionnaire that included measures pertaining to demographic and educational characteristics and relevant employment factors (e.g., duration and department of

employment) followed by a phishing task and three brief measures that assess relevant psychosocial variables and employment satisfaction. In total, the experiment took approximately 25 minutes.

Measures

Phishing discrimination task.

The phishing task comprised of 46 items, of which 23 are legitimate emails and 23 are phishes, and the presentation order of which is randomised. Each item is an image that represents an email and includes the critical information a recipient can use to judge the legitimacy of an email (identified by Steves et al., 2020, as indicators by which an email's legitimacy can be determined on the basis of two dimensions, number of salient visual cues and extent to which the premise and content of the email aligns with what would be expected by an employee at an institution such as that used here). Based on this method of categorisation, 13 (56.5%) were least difficult, 10 (43.5%) were moderately difficult, and none in our sample were categorised as most difficult to identify as phishes. Figure 2 is an example of one such phish item. Both the legitimate and phish items were sourced from the industry partner to ensure socio-geographic and contemporaneous validity to participants.

Figure 2

An example of a phishing item.

From: myGov <refund-nz8m6c@pamx1.hotmail.com.gmwebsite.com>

Sent: Saturday, 24 April 2021 9:23 AM

To: john.doe@xxxxxxx.com

Subject: Credit Note from myGov SNC0216404

CAUTION: This email originated from outside of the organisation. Do not act on instructions, click links or open attachments unless you recognise the sender and know the content is authentic and safe.

Dear myGov user,

As a result of examination we have made adjustments to your income tax return for the tax year 2020/21.

Therefore, you are eligible to receive a refund of: **\$181.79 AUD.**

To protect you from identity theft, we need to verify your identity before we process the return and issue a refund.

Please complete the identity verification process below to avoid delays in processing.

Secure Form

Important info: Your refund should be issued within 21 days of the received date. However, processing may take longer under certain circumstances.

Regards,

© The myGov Team

<http://mboxedbe71.tt.omtrdc.net/m3/testandtarget/ubox/page?mboxdefault=http://ns.spardee.com:3200/-/index.php>
Ctrl+Click to follow link

Legitimate items were safe emails sent internally and externally that contain genuine interactions or otherwise safe material. Phish items included both actual phishing attempts received and flagged by employees, and items used in previous phishing simulation campaigns which did not include the employees recruited to this study. Recipients' names and email addresses on the images were altered to either Jane Doe or John Doe but retained their organizational domain.

Upon presentation, participants were asked to observe the image and to make a judgement about whether each item was legitimate or a phish, and then asked to rate their confidence in that judgement on a scale from 0 – 100 using Qualtrics' visual analogue sliding

scale tool, anchored only by 0 and 100 with no incremental tick marks between them. There was no time limit for these items, and no feedback was given.

Overall accuracy in the task was calculated as the percentage of correct classifications across all legitimate and phish items. Confidence calibration was calculated as the difference between average confidence ratings across all items and accuracy, with negative values indicating underconfidence (confidence ratings lower than accuracy) and positive values indicating overconfidence (confidence ratings higher than accuracy). The signal detection theory parameters sensitivity (d') and decision criterion (bias) were also estimated from the number of hits (correctly identified phishing attempts) and the number of false alarms (legitimate emails incorrectly classified as phishing attempts):

$$\text{hit rate} = (\text{hits} + .5) / (\text{total number of phishing emails} + 1)$$

$$\text{false alarm rate} = (\text{false alarms} + .5) / (\text{total number of legitimate emails} + 1)$$

$$d' = z(\text{hit rate}) - z(\text{false alarm rate})$$

$$\text{bias} = -.5[z(\text{hit rate}) + z(\text{false alarm rate})]$$

Note in this context negative bias values indicate a tendency to classify all emails as legitimate.

New General Self Efficacy Scale

Self-efficacy was measured using Chen et al.'s (2010) New General Self-Efficacy Scale. The scale has 8 items that assess the extent to which a person perceives themselves to be competent, and that that competence contributes to effective positive outcomes. The items are statements (e.g., *'In general, I think that I can obtain outcomes that are important to me'*, *'Even when things are tough, I can perform quite well'*) to which respondents rate their level of agreement on a 5-point Likert scale from *'Strongly Agree'* to *'Strongly Disagree'*. The outcome measure is the average of the responses to each item. This scale has well-established psychometric properties (Chen et al., 2010).

Employee satisfaction and loyalty.

To measure employee satisfaction and loyalty, we used the Employee Satisfaction and Employee Loyalty subscales of Jun et al.'s (2006) scale designed to measure broad employee satisfaction. These two subscales each have four items, which are statements (e.g., '*I plan to make this company my career*', reflecting employee loyalty, and '*I feel personal satisfaction when I do my job well*', reflecting employee satisfaction) to which respondents rate their level of agreement on a 5-point Likert scale from '*Strongly Agree*' to '*Strongly Disagree*'. We summed the scores of the two subscales in our analyses. This measure has been validated for use in a number of organizations, industries, and nations, and has demonstrated robust psychometric properties (Jun et al., 2006; see also Kooij et al., 2010).

Propensity to Morally Disengage Scale.

We used Moore et al.'s (2012) Propensity to Morally Disengage scale, which measures each of the eight strategies used to morally disengage proposed by Bandura (1996). This measure has 24 items with three items loading onto each of the eight factors representing strategies of moral disengagement (Moral justification, Euphemistic labelling, Advantageous comparison, Displacement of responsibility, Diffusion of responsibility, Distorting consequences, Attribution of blame, and Dehumanisation). Each subscale contains three statements to which respondents rate their level of agreement on a 7-point Likert scale from '*Strongly Disagree*' to '*Strongly Agree*'. The outcome measure is the average of those three responses. We summed the scores of the eight subscales to obtain a total measure. This measure has been validated for use in adults in organizational contexts and has well-established psychometric properties (Moore et al., 2012).

Results

We first tested for differences between the five groups in the demographic variables, the phishing discrimination task performance measures, and the self-report measures (see

Table 1 and Figure 3). There were significant differences in age and number of years of employment, as the *click/click* group was somewhat older, and all initial campaign clickers had spent fewer years in employment at the organization¹. The overall accuracy in the phishing discrimination task was 82%, suggesting the absence of ceiling or floor effects on performance, and there was an overall tendency for participants to be slightly overconfident in their discrimination ability. The signal detection analysis also revealed negative bias scores, indicating that there was an overall expectation for emails to be legitimate rather than phishing attempts. It is possible that this reflects the unrealistic high proportion of phishing email examples in the task relative to the much lower likelihood of a phishing attempt in reality. There was a significant difference in accuracy among the five groups, with the *click/nothing* and *click/click* groups having the lowest accuracy, though this was not reflected in the d' scores. There was also a significant difference in bias, with the same two groups also exhibiting a stronger tendency to assume that emails are legitimate. Finally, analyses of the self-report measures revealed significant differences in employee satisfaction and loyalty, with the *click/click* group scoring the lowest on this measure. The analysis of the general self-efficacy scores was not significant, though there was a trend for somewhat lower scores in the *click/click* group.

Table 1

Demographics, Phishing Detection Task Performance and Self-Report Measures

	Behavior (initial/follow-up campaign)					<i>p</i>
	report/-	nothing/-	click/report	click/nothing	click/click	
Demographic variables						
Age	43.8 (8.7)	43.0 (10.8)	40.7 (9.9)	36.0 (9.4)	48.1 (19.2)	.022
Gender (F:M:unreported)	26:36:2	13:12:0	13:10:0	11:8:0	2:5:0	.471
Education years	15.4 (2.7)	15.7 (2.7)	15.9 (3.1)	16.0 (2.7)	15.6 (4.7)	.912

Years of employment at the organization	10.2 (8.6)	12.2 (9.9)	7.8 (7.5)	5.4 (7.3)	4.6 (4.5)	.027
Task performance measures						
Accuracy	82.9 (8.7)	81.4 (8.8)	84.0 (7.0)	78.2 (9.5)	74.5 (10.3)	.033
Overconfidence	4.8 (12.0)	6.3 (13.7)	1.1 (11.3)	9.5 (16.7)	15.0 (18.4)	.094
d'	2.09 (.60)	1.98 (.46)	2.09 (.62)	1.81 (.53)	1.72 (.41)	.201
Bias	-.59 (.11)	-.59 (.13)	-.54 (.08)	-.60 (.16)	-.70 (.13)	.021

Self-report measures						
Employee satisfaction and loyalty	14.6 (4.9)	12.4 (4.7)	15.3 (7.2)	11.1 (3.2)	8.6 (1.1)	.002
Propensity to morally disengage	137.9 (15.2)	142.9 (16.1)	133.1 (20.2)	134.3 (21.0)	146.0 (19.0)	.241
General self- efficacy	13.2 (3.6)	12.3 (4.0)	13.1 (4.3)	11.2 (2.7)	9.6 (1.7)	.057

Note. Mean and standard deviations (in parentheses) are reported for each group except for

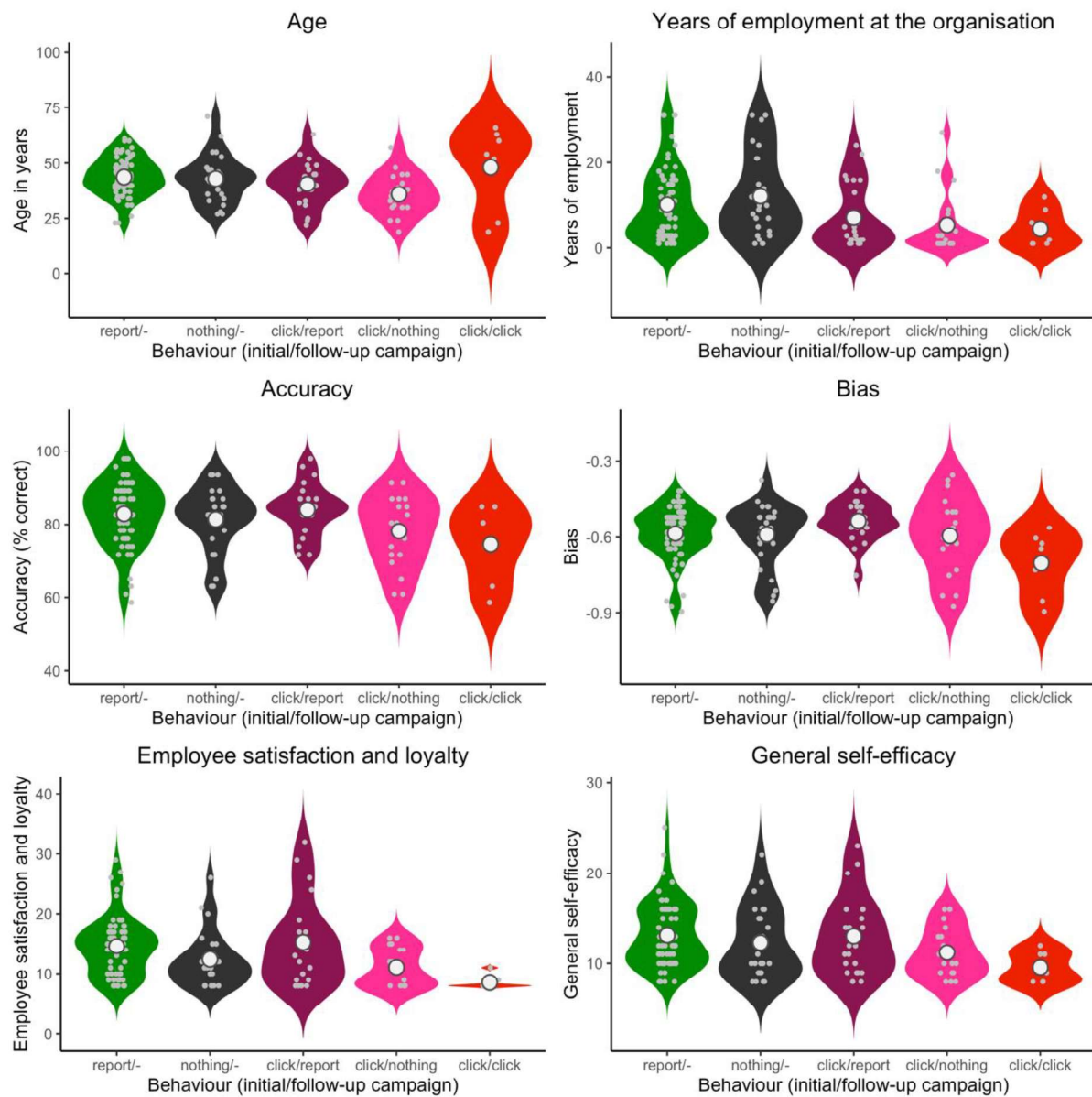
gender. The last column displays *p*-values for a chi-square test in the case of gender

(excluding the two individuals who did not report their gender) and one-way ANOVAs

comparing the five groups in the case of all other variables.

Figure 3

The Most Notable Group Differences



Although the comparisons reported in Table 1 suggest that the groups differ in several respects, we were interested in testing the extent to which each of these variables could predict behavior in the phishing simulation while controlling for the other potential predictors. We therefore performed an ordinal logistic regression to determine which factors are most predictive of behavior in the phishing simulation. The severity of the behavior in the

initial and follow up campaigns was ranked in the following way: reporting, doing nothing, and clicking were first ranked as increasingly undesirable behavior in the initial campaign, and this ranking was repeated for clickers who were followed up. This yielded the following ranking order: *report/-* (0), *nothing/-* (1), *click/report* (2), *click/nothing* (3), and *click/click* (4). The following predictors were included in a model that predicted the severity of this undesirable behavior in the phishing simulation: age, gender, number of years of education, number of years of employment at the organization, accuracy in the phishing discrimination task, bias, overconfidence, employee satisfaction and loyalty scores, propensity to morally disengage scores, and general self-efficacy scores. Note that *d'* was omitted from the list of predictors because it was highly correlated with accuracy ($r = .88$). We used the function *polr* from the R package *MASS* (Venables & Ripley, 2002) to run the model, and the function *PseudoR2* from the package *DescTools* (Andri et al., 2021) to estimate the pseudo R^2 values. We also ran a dominance analysis to rank the relative importance of the predictors using the function *domin* from the package *domir* (Luchman, 2021).

Compared to an intercept-only model, the full model afforded a significant improvement in fit (AIC = 356.4 vs 350.3 for the intercept-only vs full model; $\chi^2(10) = 26.1$, $p = .004$; McFadden $R^2 = .075$; Nagelkerke $R^2 = .200$; CoxSnell $R^2 = .187$). A summary of the model results and of the dominance analysis is presented in Table 2, in which predictors are ranked according to their relative importance, and Figure 4 illustrates the standardised dominance statistic of each predictor. The two most important predictors of suboptimal behavior in the phishing simulation were lower employee satisfaction and loyalty and fewer number of years of employment. Lower accuracy in the phishing detection task and lower reported self-efficacy also predicted suboptimal behavior to some extent, whereas the relative importance of the remaining predictors was negligible. In order to test how many of the most important predictors improve predictive accuracy, we iteratively added each predictor to the

model in order of relative importance and tested whether the inclusion of each predictor significantly improved the model fit. The inclusion of employee satisfaction and loyalty and of years of employment each significantly improved model fit ($\chi^2(1) = 9.54, p = .002$ and $\chi^2(1) = 6.83, p = .009$, respectively). The addition of accuracy did not quite significantly improve the fit ($\chi^2(1) = 3.33, p = .068$), and the further addition of self-efficacy did not significantly improve the fit ($\chi^2(1) = 1.59, p = .207$). The first two predictors are therefore significantly providing independent contributions to the predictive accuracy of the model, and the further contribution of the third and subsequent predictors is only modest.

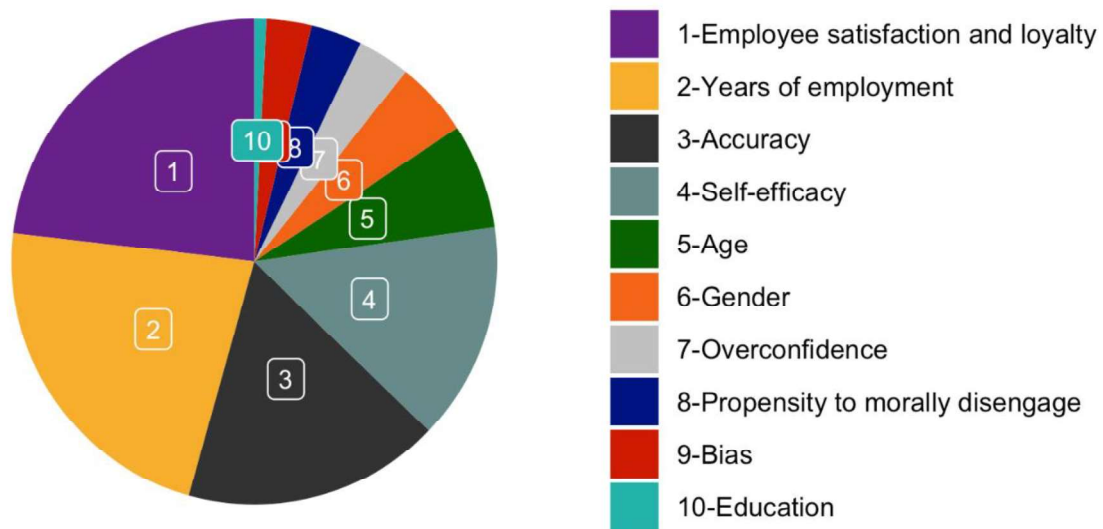
Table 2

Ordinal Logistic Regression and Dominance Analysis Results

Predictor	Ordinal regression			Dominance analysis	
	Coefficient	Standard error	<i>p</i>	Dominance rank	Standardised statistic
Employee satisfaction and loyalty	-.076	.043	.081	1	.231
Years of employment at the organization	-.054	.025	.030	2	.225
Accuracy	-.068	.036	.059	3	.171
General self-efficacy	-.082	.057	.150	4	.145
Age	-.008	.021	.709	5	.070
Gender (man)	-.492	.378	.193	6	.050
Overconfidence	-.007	.020	.720	7	.035
Propensity to morally disengage	-.017	.012	.155	8	.034
Bias	2.31	2.34	.325	9	.031
Education years	.020	.070	.290	10	.008

Figure 4

Standardised Dominance Statistic for each Predictor in Order of Relative Importance



Discussion

We investigated potential factors that might predict behavior in a naturalistic phishing simulation. We targeted individuals employed by a large organization who had been recently phished in a simulation and whose behavior had been recorded. Such simulations have excellent ecological validity as they closely resemble real phishing attempts that employees are regularly exposed to, and they give a more accurate estimate of the likelihood that employees would fall for real phishing attempts. However, such simulations provide limited information as to why some employees click on phishing links whereas others report these emails as suspicious, and why others do neither. It is commonly assumed that behavior in a phishing simulation is strongly related to the ability to discriminate phishing from legitimate emails (e.g., employees who click on phishing links might have lower discrimination accuracy), and training usually aims to improve this ability. Here, we tested this hypothesis by asking employees to complete a phishing discrimination task that required them to discriminate phishing and legitimate emails. But we also investigated whether other potential

1 factors might better predict this behavior, including motivational factors that are known to
2 influence the extent to which individuals are cautious, or motivated to report a perceived
3 threat to protect the organization. We found that although discrimination accuracy had a
4 modest effect, employee satisfaction and loyalty was a much more reliable predictor of
5 behavior in the phishing simulation. Furthermore, fewer number of years of employment at
6 the organization also significantly predicted suboptimal behavior in the phishing simulation
7 (even when controlling for the potential confounding effect of age), suggesting that
8 inexperience with the organization processes, including typical email communication, is a
9 risk factor for phishing susceptibility.
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22 Only a few other studies have previously investigated individual differences in
23 phishing susceptibility in a realistic environment. Among them, Greene et al. (2018) followed
24 up ‘clickers’ and ‘non-clickers’ after a phishing simulation and asked participants to reflect
25 on the reasons why they failed to or successfully detected the simulated phishing attempt.
26 They found that ‘clickers’ reported that they were most worried about the consequences of
27 not performing their employment duties, whereas ‘non-clickers’ were more concerned with
28 the consequences of potential cybersecurity threats. This suggests that behavior depends on
29 the individual’s perception of the consequences of the different outcomes (i.e., the relative
30 cost of failing to click on a legitimate link vs falling for a phishing attack). This study
31 suggests that even the behavior of clickers is motivated by performing well in one’s job.
32 However, it is worth noting that these retrospective self-reports may reflect after-the-fact
33 justifications for one’s behavior that perhaps did not play an important role at the time when
34 individuals made their decision. Our finding that individuals who click repeatedly report low
35 job satisfaction and loyalty to the organization suggests that at least this subgroup of
36 employees is perhaps more likely to have behaved in this way due to carelessness rather than
37 motivation to perform well. Nevertheless, it is quite possible that the other click groups,
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1 especially the *click/report* group, who reported relatively high employment satisfaction and
2 loyalty, might have fallen for the first phishing attempt because of higher concern with
3 performing their duties.
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7 Another study by Boritz et al. (2019) also followed up recently phished employees
8 and found that while gender and education levels were not associated with phishing behavior,
9 more years of work experience, lower self-reported scepticism and suspicion of hostility and
10 somewhat higher interpersonal trust predicted phishing susceptibility, and that these
11 relationships might be moderated by risk propensity and cognitive inhibitory control.
12 Consistent with this, Workman et al. (2008) also found that higher trust predicts increased
13 phishing susceptibility. He also explored additional individual differences and found phishing
14 susceptibility to be associated with higher social commitment (an attitude that causes people
15 to perform perseverant actions), higher obedience and older age. He did, however, find that
16 phishing susceptibility was associated with higher education levels. Moreover, Workman
17 reported that phishing susceptibility was unrelated to reactance (a tendency to oppose
18 authority) and previous experience with falling for phishing attack.
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36 Furthermore, a study by Broadhurst and colleagues (Broadhurst et al., 2019) found
37 that neither cybercrime awareness, IT competence, or perceived Internet safety predicted
38 phishing susceptibility. Finally, a recent large study by Lain et al. (2021) investigated
39 demographic factors associated with phishing susceptibility in a series of phishing
40 simulations in a large organization. They found that both younger and older employees were
41 more susceptible to fall for simulated phishing attacks, but that gender was not associated
42 with susceptibility. Lain et al. further followed up some of the employees who had taken part
43 in the simulation and found that approximately 40% of those who clicked and received
44 immediate feedback and training (a protocol similar to that used in our study) reported that
45 the training webpage made them feel safer and that they believed the company protects them
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1 from phishing attempts. The authors proposed that these beliefs might explain their
2 counterintuitive finding that immediate feedback and training was associated with a higher
3 re-click rate (perhaps potentially explaining some of the re-clicking behavior we observed),
4 although they did not directly test this hypothesis by comparing the behavior of those who
5 reported feeling safer after seeing the feedback page and those who did not.
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11 In summary, we replicated some of the previously reported effects though some
12 inconsistencies remain. We found that suboptimal behavior in the phishing simulation was
13 associated with older age, which is consistent with the findings of Lain et al. (2021) and
14 Workman et al. (2008), and like Lain et al. (2021) and Broadhurst et al. (2019), we found that
15 gender did not predict phishing behavior. We also found that education did not predict
16 phishing susceptibility, consistent with Boritz et al. (2019), though inconsistent with the
17 results reported by Workman (2008). Interestingly, while we found phishing susceptibility to
18 be associated with fewer years at the organization, Boritz et al. (2019) reported that more
19 work experience (not organization-specific) was associated with increased phishing
20 susceptibility.
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36 We further found that clicking behavior was associated with a more negative bias,
37 suggesting that those who clicked had a higher tendency to assume emails are legitimate
38 rather than phishing attempts. Although we measured a different construct, this result is
39 somewhat consistent with Boritz et al.'s (2019) and Workman's (2008) finding that clicking
40 behavior is associated with greater trust. These findings, are however, somewhat inconsistent
41 with Broadhurst et al.'s (2019) finding that phishing susceptibility is unrelated to perceived
42 Internet safety if one considers it as a form of trust.
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53 Our study has revealed, however, other factors that were more predictive of
54 cybersecurity behavior than those mentioned above. We investigated previously unexplored
55 factors, including phishing detection accuracy, which requires administering a task that
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1 includes both legitimate and phishing examples to accurately estimate discrimination ability.
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3 Furthermore, we have captured a wider range of behavior in the phishing simulation than
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5 previous studies, including neither reporting or clicking on the link, and behavior in the
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7 follow-up campaign, which has allowed us to further subdivide the group of clickers
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9 according to their responses to the second phishing attempt. We found that, surprisingly,
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11 phishing detection accuracy is not the most informative predictor of behavior in the phishing
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13 simulation. This suggests that although further training to identify phishing attempts might
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15 benefit some employees, other types of intervention might be required to improve the
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17 cybersecurity behavior of a wider range of employees. In particular, our results suggest that
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19 improving job satisfaction and loyalty to the organization might have widespread benefits
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21 that include more cautious cybersecurity behavior. Finally, our study has also revealed that
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23 the second critical factor that predicts suboptimal behavior is job inexperience, suggesting
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25 that targeted cybersecurity training for newer employees might benefit the organization.
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36 Footnote:

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39 ¹ Note that we do not report statistics for the numerous possible post-hoc comparisons as the
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41 subsequent analysis ranks the behavior in the phishing simulation according to its
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43 undesirability and aims to identify the most reliable predictors of this behavior among the
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45 factors investigated.
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Declaration of interests

☐The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Nathan Beu reports financial support was provided by Centre for Research on Engineering Software Technologies. Asangi Jayatilaka reports financial support was provided by Centre for Research on Engineering Software Technologies.